

Vector competence of Belgian *Culex pipiens* for West Nile virus under different temperature conditions

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Introduction

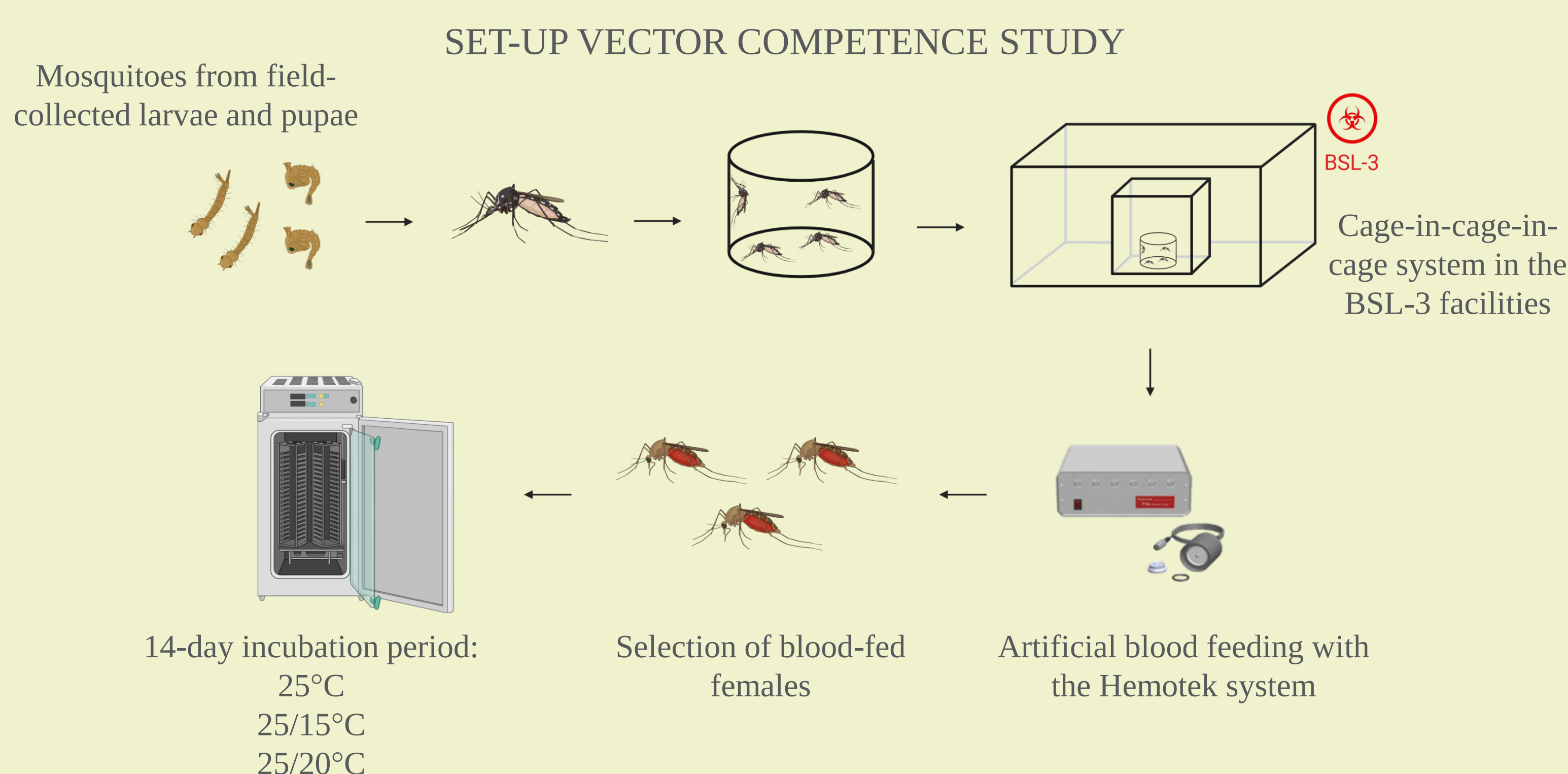
- Culex pipiens* is the **most abundant mosquito species** in Belgium
- Proven competent** for West Nile virus (WNV) in **neighboring countries**
 - Considered to be an **important vector** for **West Nile virus**
- Vector competence** in Belgian mosquitoes has been **little investigated**
- Previous research showed a **transmission efficiency of 4,3%¹**
- Fluctuations in temperature** due to climate change can lead to differences in vector competence

Questions

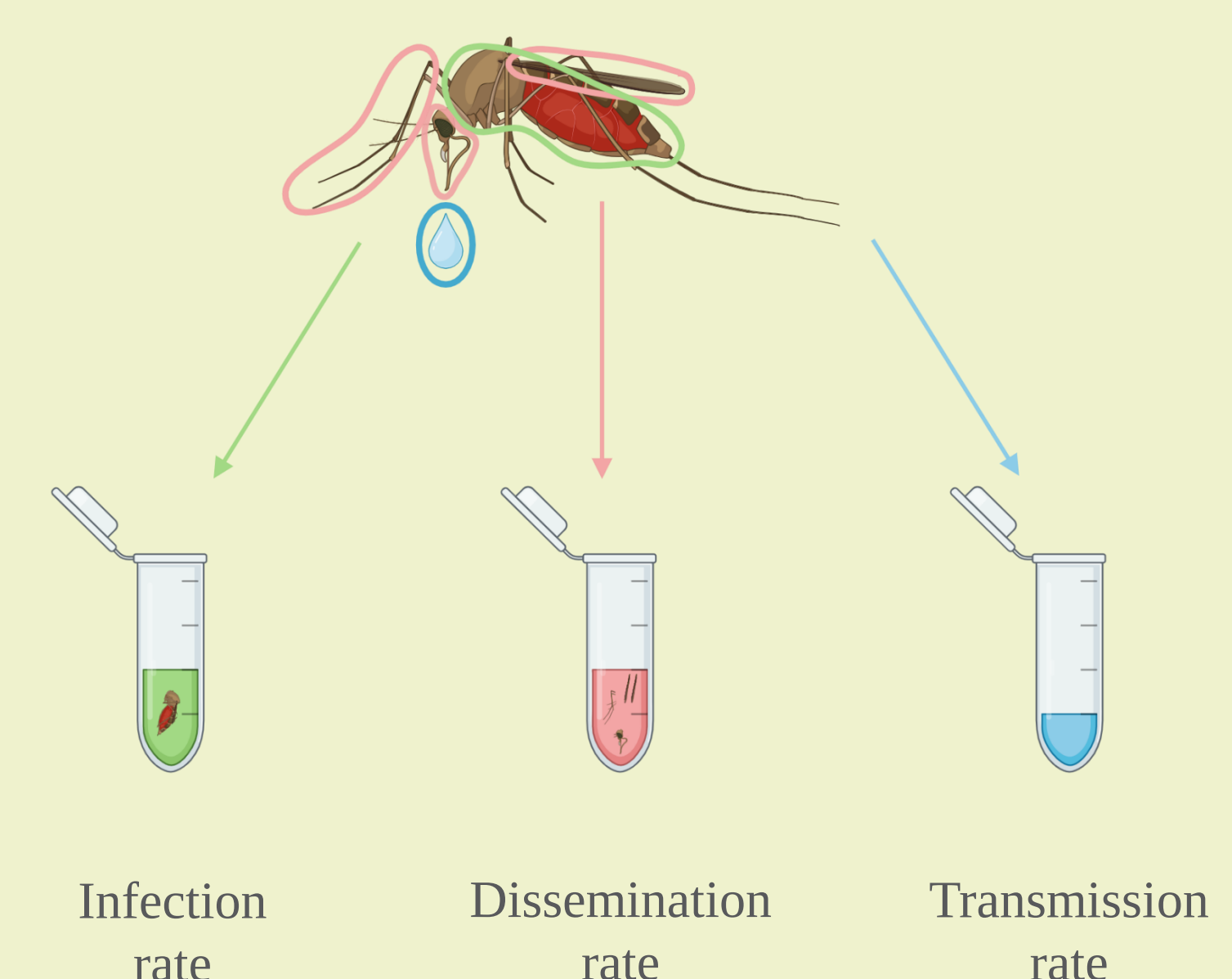
ARE BELGIAN *CULEX PIPPIENS* MOSQUITOES **COMPETENT VECTORS** FOR WEST NILE VIRUS?

HOW DO **TEMPERATURE CONDITIONS** AFFECT VECTOR COMPETENCE?

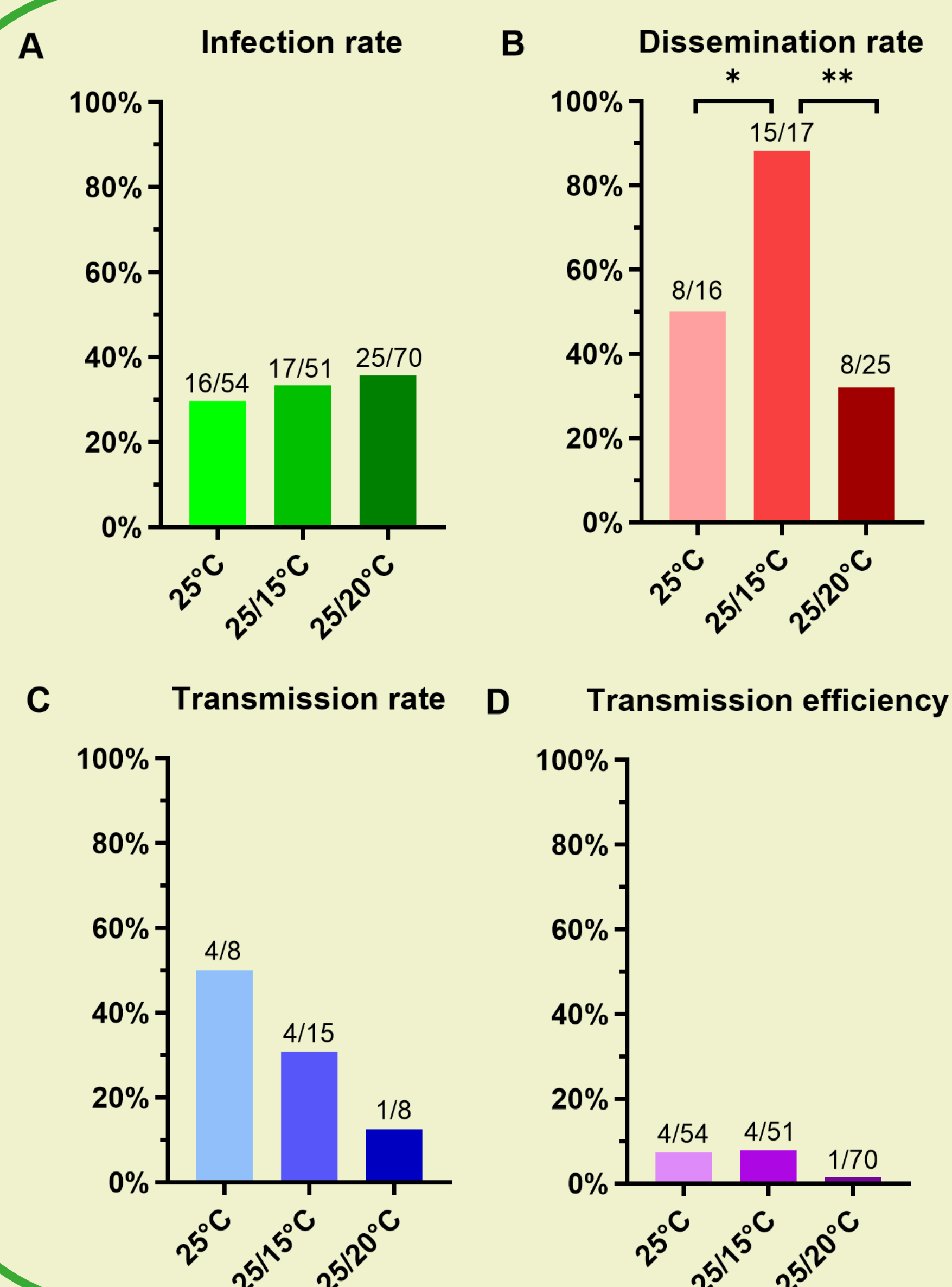
Methods



SAMPLE COLLECTION AFTER 14-DAY INCUBATION PERIOD



Results



At 25°C, a total of 29.6% of mosquitoes was positive for WNV in the abdomens, while the infection rate at 25/15°C and 25/20°C was 33.3% and 35.7%, respectively (panel A).

A dissemination rate of 50% at 25°C was obtained, which was found to be statistically lower than that at 25/15°C (88.2%). At 25/20°C, a dissemination rate of 32% was obtained, which was also statistically lower in comparison with the 25/15°C temperature condition (panel B).

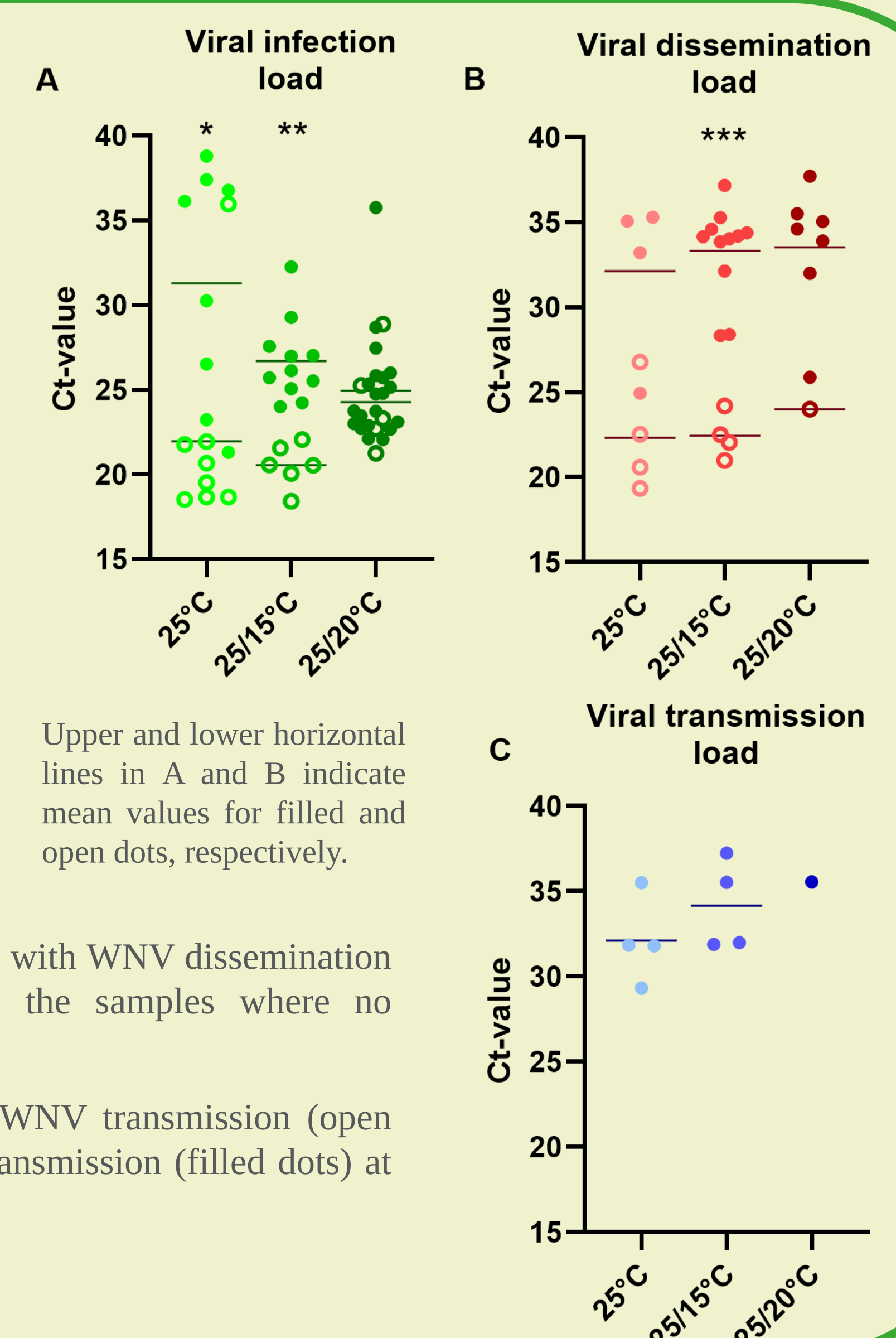
A total of 50% at 25°C was positive for WNV in the saliva. At 25/15°C, the transmission rate amounted to 30.8%, while the rate corresponded to 12.5% at 25/20°C (panel C). These rates lead to comparable transmission efficiencies of 7.4%, 7.8%, and 1.4% at 25°C, 25/15°C, and 25/20°C (panel D).

* p = 0.0491; ** p = 0.0005

At 25°C and 25/15°C, the viral loads of the infection samples with WNV dissemination (open dots) were significantly higher than compared to the samples where no dissemination (filled dots) was found (panel A).

In addition, the viral dissemination loads of samples with WNV transmission (open dots) were significantly higher compared to those without transmission (filled dots) at 25/15°C (panel B).

* p = 0.0045; ** p = 0.0002; *** p = 0.0015



Conclusions

- Belgian *Culex pipiens* mosquitoes are **competent for West Nile virus** at 25°C, 25/15°C, and 25/20°C
- Vector competence is **not influenced by temperature** during the extrinsic incubation period
- The **transmission efficiencies** of 7.4% (25°C), 7.8% (25/15°C), and 1.4% (25/20°C) reported here **are in line with the previously reported data¹**
- High viral loads facilitate the dissemination and transmission** of WNV
 - Important factors **driving the spread** of WNV in the mosquito's body
- In contrast to Japanese encephalitis virus, **WNV transmission** in *Culex pipiens* is **not inhibited** at 25/15°C²
 - **Different capability** of both viruses to overcome mosquito barriers

References

- ¹Soto et al., "Belgian *Culex Pipiens Pipiens* Are Competent Vectors for West Nile Virus While *Culex Modestus* Are Competent Vectors for Usutu Virus."
- ²Van den Eynde et al., "Relevant Day/Night Temperatures Simulating Belgian Summer Conditions Reduce Japanese Encephalitis Virus Dissemination and Transmission in Belgian Field-Collected *Culex Pipiens* Mosquitoes."